

COM4532 – Final Report

You are required to submit a final report describing the project you proposed. The report must clearly explain the problem, the proposed solution, the implementation details, and the obtained results. The report must consist of **at least 15 pages** (excluding references and appendices).

The report should contain the following sections.

1. Introduction and Background (at least 2 pages)

Provide an overview of the problem addressed by your project.

The section should include:

- Description of the problem
- Motivation
- Current approaches and limitations
- Why blockchain technology is relevant to the problem

For data-driven projects, also briefly introduce the blockchain ecosystem and data source used in the study.

2. Related Work and Technical Background (at least 2 pages)

Provide the necessary technical background and summarize related studies.

Topics may include:

- Blockchain fundamentals
- Nodes
- Transactions
- Consensus mechanisms
- Smart contracts
- DeFi concepts
- Blockchain analytics
- Machine learning concepts

The content should be selected according to the project topic.

3. System Design / Problem Construction (at least 4 pages)

Describe the design of your solution.

For Platform-Based Projects

The section should answer the following questions:

- Is the platform built on a permissioned or permissionless blockchain? Why?
- What are the entities and roles in the system?
- How are users registered?
- How are smart contracts used?
- How is data stored?
- How does the consensus mechanism support the platform?
- How are security and privacy concerns addressed?

Include architecture diagrams whenever possible.

For Smart Contract Projects

The section should answer the following questions:

- What problem does the smart contract solve?
- What are the actors interacting with the contract?
- What are the major contract functions?
- What are the access control mechanisms?
- What security considerations were taken into account?
- How does the contract interact with blockchain assets or tokens?

Include contract design diagrams and interaction flows whenever possible.

For Blockchain Data Analytics / AI Projects

The section should answer the following questions:

- What is the analysis task?
 - Classification
 - Clustering
 - Anomaly Detection
 - Prediction
- What blockchain data is used?
- What are the data sources?
- What entities are analyzed?
 - Wallets
 - Transactions
 - Smart Contracts
 - Tokens
- What features are extracted?
- What are the expected outputs?

Include data flow diagrams whenever possible.

4. Implementation (at least 3 pages)

Describe the implementation details.

Platform-Based Projects

Include:

- Smart contract implementation
- Frontend/backend components
- Frameworks and tools used
- Screenshots

Smart Contract Projects

Include:

- Contract structure
- Major functions
- Deployment details
- Testing procedures

Blockchain Data Analytics / AI Projects

Include:

- Data collection process
- Preprocessing
- Feature engineering
- Machine learning / deep learning models
- Hyperparameters
- Software libraries used

Provide code snippets where appropriate.

5. Evaluation and Results (at least 2 pages)

Demonstrate that your solution works.

Platform-Based Projects

Examples:

- Functional testing
- User scenarios
- Security analysis
- Gas consumption analysis

Smart Contract Projects

Examples:

- Unit tests
- Security testing
- Gas efficiency analysis
- Contract verification results

Blockchain Data Analytics / AI Projects

Examples:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Include tables and figures whenever possible.

6. Discussion and Future Work (at least 1 page)

Discuss:

- Strengths of your solution
- Limitations
- Potential improvements
- Future extensions

7. References

Provide all references used in the report.

Use a consistent citation format.

Important Notes

- The report must clearly demonstrate your own work.
- Include figures, architecture diagrams, screenshots, tables, and experimental results whenever possible.
- For AI-based projects, simply training a model is not sufficient; feature selection, evaluation methodology, and interpretation of results should also be discussed.
- For smart contract projects, security considerations and testing are expected.
- For platform projects, architecture and blockchain integration should be clearly explained.
- Reports copied verbatim from the Internet, AI tools, or other sources will receive no credit.

Due date for the project report is: **June 21 till 11.59 pm.**